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A search-based theory of merger waves☆,☆☆

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ABSTRACT

We develop a search-based theory of mergers and acquisitions (M&A) with heterogeneous firms and endogenous search complementarities. We use this model to study how merger incentives, endogenous innovation, and the firm size distribution interact. We derive the law of motion of the firm size distribution, provide closed-form solutions, and solve for firms' investment in internal innovation and external acquisitions. Strategic complementarities in acquisition incentives can generate multiple equilibria, including an unstable equilibrium with high M&A activity. The model therefore offers a new perspective on merger waves as episodes in which favorable fundamentals and coordination interact to generate temporary bursts of acquisition activity. Competition policy can affect both the level of equilibrium merger activity and the set of equilibria the economy can sustain, with implications for innovation, markups, and welfare.

> "A successful search program combines methodical hard work with occasional instances of pure luck; that is, 'being in the right place at the right time.' But luck in the M&A business isn't just happenstance. Being in the right place at the right time is the result of considerable effort."
>
> [Jeffrey C. Hooke, M&A: A Practical Guide to Doing the Deal (2015, P. 47)]

1. Introduction

Mergers and acquisitions (M&A) remain a central force shaping firm dynamics, market structure, and the allocation of innovative activity. When made by large incumbents, acquisitions mechanically increase concentration at least temporarily, and this has intensified concerns that merger waves may translate into higher markups and weaker competition.[2] At the same time, M&As can reallocate assets to more productive firms, generate synergies, and enhance product qualities. These opposing forces are especially salient in industries where firms can expand both through acquisitions and through in-house research and development (R&D).[3]

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[2] See for instance the former FTC declarations on start-ups https://techcrunch.com/2024/06/15/ftc-chair-lina-khan-on-startups-scaling-and-innovations-in-potential-law-breaking/ for the United States and the new merger control guidelines issued in the United Kingdom https://www.skadden.com/insights/publications/2024/05/uk-revamps-merger-control

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Understanding how firms allocate resources across these two margins, and how that allocation feeds back into market power and welfare, is therefore central to the design of antitrust policy.

Most of the existing literature on merger policy is conducted in a static or partial equilibrium framework, limiting the ability to analyze the dynamic interaction of merger incentives, innovation choices, and the endogenous distribution of firm productivities. Yet these interactions can be of first order importance. Searching for acquisition targets is costly and the return to search depends on the distribution of available firms, which is itself shaped by past M&A activity and firms' innovation decisions. At the same time, the private return to innovation depends on the expected path of market power and on whether firms anticipate growing through internal R&D or external acquisitions. A consistent evaluation of merger policy therefore requires to consider the dynamic environment in which M&A, innovation, and market power are jointly determined.

In this paper, we develop a continuous-time search-based theory of mergers and acquisitions with heterogeneous firms, endogenous markups, and endogenous innovation. Firms are heterogeneous in the quality of their products, which they can improve in two ways: by completing successful R&D projects that raise productivity incrementally, or by merging with another firm, which combines the two firms' productivities augmented by potential synergy gains. Product-market power arises endogenously in this environment from bilateral bargaining between firms and consumers. As in recent search-theoretic models of decentralized trade, firms devote resources in rent-seeking activities in order to improve their bargaining position, resulting in equilibrium markups that are endogenous and heterogeneous across firms (Choi and Rocheteau, 2024). Crucially, in our model, rent-seeking costs are not constant but scale with the average product quality in the economy. This links a firm's incentive to raise its markup to the productivity of its competitors and delivers a market-power specification in which other firms' quality matters for equilibrium markups. We then embed firms' endogenous M&A and R&D intensity decisions into this frictional goods-market framework with endogenous market power.

Our first contribution is analytical. We characterize the law of motion for the endogenous distribution of firm product quality in an environment with firm exit, successful R&D, and mergers. Using characteristic-function methods, we solve the Kolmogorov Forward Equation (KFE) in closed form and derive recursive expressions for all the moments of the stationary distribution. In particular, we show that the first two moments of firm product quality are strictly increasing in the M&A matching rate, M&A synergies, the R&D success rate, and the post-innovation productivity jump. These results make transparent how acquisitions and innovation jointly shape the firm distribution.[4]

Next, we embed this firm-dynamics block into a tractable model of endogenous market power. Firms bargain bilaterally with buyers in decentralized product markets. A firm's product quality raises the gains from trade in a match and therefore raises its incentive to invest in rent seeking. This implies that more productive firms command higher bargaining power and charge higher markups. We solve the firm's dynamic problem in closed form and obtain explicit expressions for equilibrium markups, market shares, concentration, and firm values. Because the cost of rent seeking increases with average quality in the economy, our framework captures the idea that market power is harder to sustain when rivals are more productive. In this sense, our adaptation of Choi and Rocheteau (2024) provides a tractable way to capture markups that depend not only on a firm's own productivity, but also on the broader competitive environment.

Our second contribution is to show that the tradeoff between M&A search and innovation is non-trivial and that multiple equilibria can coexist. We endogenize firms' resource allocation across M&A search and innovation by letting entrants choose how to split a fixed resource/effort endowment between M&A and R&D intensities. This introduces a natural trade-off between external growth and internal growth. We show that the game among firms can exhibit strategic complementarities in M&A effort: when other firms search more intensely for acquisitions, the distribution of future merger opportunities improves, which can increase an individual firm's incentive to do the same. Under empirically relevant parameter values, this mechanism generates multiple equilibria. In particular, a stable equilibrium with relatively low M&A and high R&D can coexist with an unstable equilibrium featuring high M&A and low R&D. Thus, the same fundamentals may be consistent with very different industry configurations depending on how firms coordinate their M&A and R&D intensity choices.

These multiple equilibria have important implications for welfare analysis in connection with market power, concentration, and product quality. By using a calibrated numerical example, we show that the stable low-M&A equilibrium is typically associated with higher R&D intensity, higher average product quality, and lower concentration, but also higher average markups. Consumer welfare is also higher in this equilibrium, because the gain from better product quality dominates the adverse effect of higher markups. More broadly, the model implies that consumers face a trade-off between product quality and markups across equilibria. This result relates to recent work emphasizing that markup movements need not map cleanly into welfare changes (Menzio, 2026) and cannot serve as a sufficient statistic for welfare analysis. In our environment, while markups are not themselves efficient, the ability of firms to charge markups strengthens incentives to invest in activities that improve future product quality. As a result, an equilibrium with higher average markups but lower concentration can also deliver higher consumer surplus and higher social welfare.

Our numerical analysis yields several additional lessons. First, average markup and concentration as measured by the Herfindahl–Hirschman Index (HHI) need not move in tandem. In our calibration, the unstable equilibrium with higher M&A and lower R&D features lower average markups but higher HHI, because lower average quality makes rent-seeking activities cheaper and amplifies

[3] News coverage during the 2003–2009 merger wave frequently emphasized that firms were using acquisitions to grow by "buying" innovation rather than investing in risky R&D to develop frontier technologies (Forbes, 2008; Bloomberg, 2008).

[4] To our knowledge, the only other firm-dynamics models that incorporate both M&A and R&D are Cavenaile et al. (2021) and Celik et al. (2022), both of which only solve numerically for the endogenous distribution.

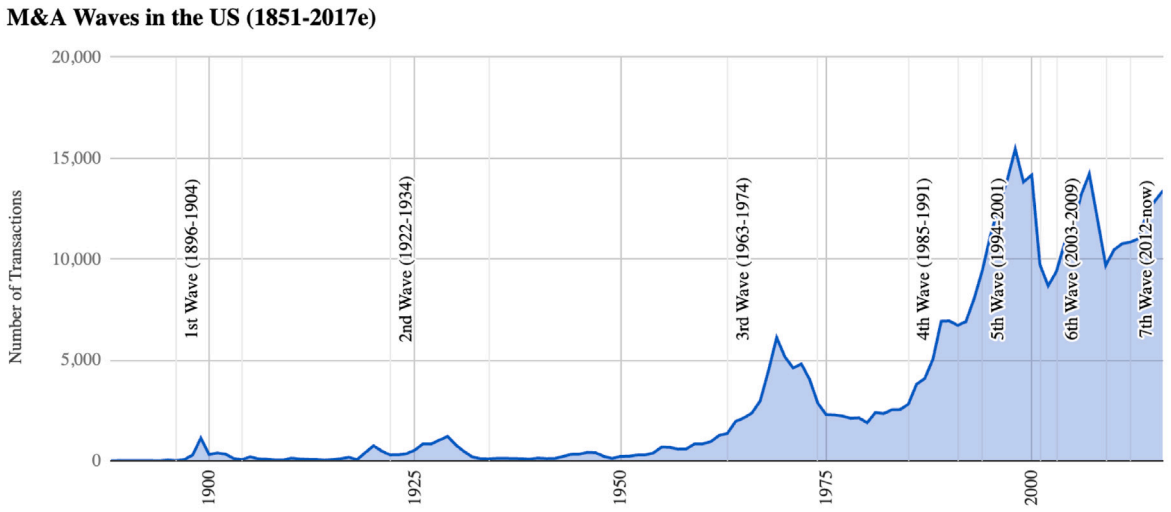


Fig. 1. M&A waves in the US (1851–2017). *Source:* https://imaa-institute.org/m-and-a-us-united-states/ (accessed on 20 March 2026).

the sales concentration of the most productive firms in equilibrium. Second, the constrained planner would like firms to devote less resources to M&A search and more to R&D efforts than they do in either private equilibrium. The reason is that strategic complementarities in acquisition activity crowd out innovation. Hence, even when acquisitions generate synergies, decentralized firms overinvest in M&A relative to internal innovation.

Third, and importantly, the model provides a novel perspective on merger waves. Recurring boom-bust episodes of intense M&A activity have been observed in many economies, including in the United States over the past 150 years or so (Fig. 1). We show that, merger waves need not solely reflect large changes in fundamentals but can reflect multiple equilibria resulting from strategic search complementarities. In particular, a favorable shift in financing conditions, technology, or regulations can move the economy from a low-merger regime to a temporarily elevated merger regime sustained by strategic complementarities and coordination in M&A search effort. In this interpretation, merger waves are amplified episodes in which expectations and fundamentals reinforce one another. Furthermore, we show that the high-merger equilibrium is *unstable*, and merger waves are therefore inherently fragile: any tightening of antitrust enforcement, worsening of credit conditions, or disappointing post-merger outcomes can break the coordination sustaining high acquisition activity and push the economy back to the low-merger equilibrium.

Taken together, these observations lead naturally to policy implications. The model implies that competition policies affect not only the *level* of merger activity in a given equilibrium, but also *the set of equilibria* that the economy can sustain. In our framework, the marginal cost of acquisition search is a reduced-form object that captures the stringency of merger policy. A stricter antitrust regime raises the cost of searching for and completing acquisitions, and, as a result, reduces the stable equilibrium level of M&A, but, because of strategic complementarities can have the opposite effects on the search intensities in the unstable, high-search equilibria, creating risks of abrupt corrections. Moreover, it can also eliminate the high-merger equilibrium altogether. The same logic applies to innovation policy: subsidizing R&D, by making internal growth more attractive relative to acquisitions, can effectively rule out multiple equilibria and push the economy towards the equilibrium with a higher welfare. Thus, antitrust policy and innovation policy interact through firms' endogenous resource allocation decisions, with profound welfare consequences.

Related literature. Our paper contributes to several strands of literature at the intersection of mergers and acquisitions, innovation, market power, and industry dynamics. First, our search-theoretic approach to market power provides an alternative lens to the extensive analyses of horizontal mergers and antitrust policy built on static frameworks. A classic starting point is Williamson (1968), who emphasizes the trade-off between efficiency gains and increased market power in merger analysis. Merger approval rules have since been studied in a sequence of formal models, including Besanko and Spulber (1993), Neven et al. (2005), Nocke and Whinston (2010), Ottaviani and Sotcrensen (2011), Nocke and Whinston (2013), and Guadalupe et al. (2021). Kaplow (2021) surveys this literature and highlights the shortcomings of static market concentration-based antitrust policies, including their limited ability to capture innovation, dynamic effects, and the general-equilibrium implications of mergers.[5] Our paper complements this literature by speaking to the growing recognition that antitrust policy should account for innovation and the associated dynamic

[5] Similarly, aggregate implications of antitrust policies against cartelization and collusion between firms are not well-known. Using a quantitative framework that accounts for the misallocation costs of collusion, Moreau and Panon (2023) find potentially important costs of competition distortions from cartelization.

efficiency gains. Specifically, we provide a dynamic framework in which policy affects not only realized merger activity, but also firms' incentives to innovate and the set of equilibria the economy can sustain.

Second, our paper is related to the growing literature that studies mergers and acquisitions in search-and-matching environments. Guadalupe et al. (2021) propose a static assignment model of mergers and acquisitions and study granular complementarities between targets and acquirers. Closer to our paper, David (2021) proposes a dynamic theory of M&A where search happens exogenously and firms do not have markups, therefore preventing to study the potentially anti-competitive impact of mergers. Chan and Qi (2025) extends (David, 2021) with endogenous markups and study the impact of mergers on aggregate volatility. Cavenaile et al. (2021) and Celik et al. (2022) enrich this class of models by incorporating innovation. Our main contribution relative to this literature, is to endogenize M&A search intensity and provide analytical expressions linking search intensities with market structure in equilibrium. This allows us to characterize how strategic complementarities in acquisition incentives can generate multiple equilibria, and how those equilibria differ in terms of M&A activity, innovation, markups, concentration, and welfare. In contrast, both the models of Cavenaile et al. (2021) and Celik et al. (2022) admit a unique solution because they take the M&A search intensity as given.

Third, the endogenous search complementarities in our model contribute to the literature on merger waves by offering a new perspective on why merger waves may emerge and unravel abruptly. Cho and Chung (2022), in a recent review of the empirical and theoretical literature on merger waves, note that a consensus on the causes of merger waves has remained elusive. Most studies tend to emphasize one of two broad classes of explanations. Under the managerial-behavior view, merger waves arise from factors such as agency motives, market sentiment, and the overvaluation of acquirers' equity. In contrast, according to the neoclassical view, merger waves emerge in response to capital-market conditions, triggered by exogenous industry shocks. Papers in this latter tradition include Gort (1969), Andrade et al. (2001), Lambrecht (2004), and Harford (2005), among others. While we also provide a neoclassical theory of mergers, to our knowledge, we are the first to develop a theoretical framework in which strategic complementarities in acquisition incentives can generate multiple equilibria, including an unstable equilibrium with high M&A activity consistent with the "hot" merger markets described in the merger wave literature (Harford, 2024). The model therefore offers a new perspective on merger waves as episodes in which favorable fundamentals, strategic complementarities, and coordination interact to generate temporarily elevated acquisition activity.

Fourth, our paper is also related to an active search-theoretic literature on market power which is not concerned with M&As. In particular, our search-theoretic micro-foundations for market power draw on the canonical sequential bargaining approach of Rubinstein and Wolinsky (1985), recently revived in the New Monetarist literature (Choi and Rocheteau, 2024; Kala et al., 2024). While Choi and Rocheteau (2024) take the product quality distribution as given, we make it an endogenous object by introducing M&A search dynamics. These search motives in turn generate an endogenous firm size distribution whose dynamic properties are reminiscent of the information percolation mechanism studied in Duffie et al. (2010), Duffie (2011), and Andrei and Cujean (2017). Characteristic-function techniques adapted and extended from the information percolation models allow us to analytically characterize the endogenous firm distribution.

Finally, our paper is related to an expanding macroeconomic literature studying market concentration and business dynamism. Díez et al. (2018) and De Loecker et al. (2019) document the rise of market power and business concentration, while Grullon et al. (2015) and Gutiérrez and Philippon (2017) connect rising concentration and market power to weaker competition and more permissive antitrust enforcement. Jovanovic and Rousseau (2002, 2008) emphasize the reallocative role of mergers and merger waves. Akcigit and Ates (2023) study the strategic behavior between competing firms, its effect on their innovation decisions, and the resulting "best-versus-the-rest" dynamics. Our paper contributes to this literature by providing a tractable model in which acquisitions and innovation jointly shape the endogenous distribution of firm quality, equilibrium markups, and concentration.

The rest of the paper proceeds as follows. Section 2 develops the model and characterizes the stationary distribution of firm quality, endogenous markups, firms' value functions, equilibrium M&A and R&D choices, and welfare. Section 3 presents the calibration, numerical results, and policy counterfactuals. Section 4 concludes.

2. A search-based model

This section develops our model and presents the main theoretical results. We proceed in several steps. We first characterize the equilibrium distribution of firm size and the firms' value function for a given exogenous M&A search intensity and R&D success intensity. We then endogenize these intensities and show how M&A and R&D activities responds to deep parameters and antitrust policy.

2.1. Equilibrium mergers and acquisitions dynamics with heterogeneous firms

Time is continuous and runs forever, $t \in [0, \infty)$. There are a unit-mass continuum of target firms/entrepreneurs/sellers with products of heterogeneous quality, ε. Target firms in the economy are subject to an overlapping generations structure as follows. At a Poisson rate $\eta > 0$, a target firm dies and is immediately replaced with a new-born target firm whose product quality is drawn from the probability density function (pdf) $g : \mathcal{E} = [\varepsilon_l, \varepsilon_h] \to \mathbb{R}$. Second, at a Poisson rate $R > 0$, a target firm successfully completes an R&D project, which raises the firm's productivity from ε to $\varepsilon + B$ for some $B > 0$. Third, at a Poisson rate $\lambda > 0$, a target firm is matched with another target firm to negotiate an M&A (mergers-and-acquisitions) deal. These three events — "death", "R&D success", and "matching"— are assumed to be independent from one another and i.i.d. in the cross section of firms. We follow

the finance search literature (Üslü, 2019; Farboodi et al., 2023) in assuming that, following a matching shock, a Pareto-optimal bargaining procedure with symmetric bargaining powers immediately results in a deal.

We describe and characterize the target firms' continuation utilities and surplus from M&A deals in the next section. Here we take as given the outcome of the M&A process: an M&A between two target firms with initial product qualities of ε' and ε'' gives rise to a new firm with a product quality of $A+\varepsilon'+\varepsilon''$, where $A \geq 0$ reflects the synergy benefit of M&As, and the acquired firm exits the market and is replaced by a new entrant with a product quality from the pdf g. The endogenous pdf of product quality, ε, that we name $f(\cdot)$, hence, evolves according to the following Kolmogorov Forward Equation (KFE):

$$\dot{f}(\varepsilon) = -\underbrace{(\eta+\lambda+R) f(\varepsilon)}_{gross\ \ outflow} + \underbrace{R f(\varepsilon - B)}_{R\&D} + \underbrace{\left(\eta+\frac{\lambda}{2}\right) g(\varepsilon)}_{entry}$$
$$+ \underbrace{\frac{\lambda}{2}\int_{-\infty}^{\infty} f(\varepsilon') f(\varepsilon - A - \varepsilon')\, d\varepsilon'}_{mergers}. \tag{1}$$

The first term on the RHS of (1) captures the (gross) outflow from the density of firms with product quality ε due to death, M&A meetings, and successful R&Ds. The second term captures inflow due to R&D. That is, firms with product quality $\varepsilon - B$ successfully complete an R&D project at Poisson rate R and become a firm with product quality ε. The third term captures the entry of firms with product quality ε, following the death and M&A of predecessor firms. Note that $\lambda/2$ reflects the fact that M&As occur at rate λ and after each M&A the acquired firm exists the market and is replaced by another firm with a product quality from the pdf g.[6] The last term of (1) captures the entry of firms with synergy benefit following M&As. More specifically, any firm with product quality ε' may end up with the post-M&A product quality ε, conditional on meeting a firm with initial product quality of $\varepsilon - A - \varepsilon'$. These continuum of possibilities are behind the convolution integral in the last term. Since the convolution integral complicates the computation of the pdf, we will make use of the characteristic function (Lukacs, 1970, p. 5):

$$\hat{f}(z) = \int_{-\infty}^{\infty} e^{iz\varepsilon} f(\varepsilon)\, d\varepsilon.$$

Using the characteristic function allows to solve the KFE in closed form. We have the following proposition.

Proposition 1 (*Steady State Solution*)**.** *Let $\hat{f}(\cdot)$ be the characteristic function of the endogenous distribution of product quality. Then, $\hat{f}(\cdot)$ satisfies the following KFE:*

$$\dot{\hat{f}}(z) = -\left(\eta+\lambda+R\left(1-e^{iBz}\right)\right)\hat{f}(z) + \left(\eta+\frac{\lambda}{2}\right)\hat{g}(z) + \frac{\lambda}{2}e^{iAz}\left(\hat{f}(z)\right)^2. \tag{2}$$

At the steady state, the characteristic function admits the following explicit expression:

$$\hat{f}(z) = \frac{\eta+\lambda+R\left(1-e^{iBz}\right) - \sqrt{\left(\eta+\lambda+R\left(1-e^{iBz}\right)\right)^2 - 2\lambda e^{iAz}\left(\eta+\frac{\lambda}{2}\right)\hat{g}(z)}}{\lambda e^{iAz}}. \tag{3}$$

As there is a one-to-one correspondence between the characteristic function $\hat{f}$ and the cumulative distribution function, we conclude that the KFE admits a unique solution, f, such that

$$f(\varepsilon) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\varepsilon z}\hat{f}(z)\, dz.$$

Furthermore, the characteristic function can be used to provide analytical expressions for all the moments of the solution. These analytical expressions allow us to study how deep parameters shape the steady state distribution of firms. In particular, we have the following comparative statics:

Proposition 2 (*Moments of the Steady State Distribution*)**.** *At the steady state, the first two moments of the firms' product quality distribution are strictly increasing i) in the M&A matching rate, λ, ii) in the amount of synergies, A, iii) in the R&D success rate, R, and (iv) in the post-R&D productivity jump, B.*

For $n \geq 1$, the nth moment of the stationary distribution admits the following recursive representation

$$\mathbb{E}_f\left[\varepsilon^n\right] = \frac{1}{\eta}\left(\eta+\frac{\lambda}{2}\right)\mathbb{E}_g\left[\varepsilon^n\right] + \frac{1}{\eta}R\sum_{k=1}^{n}\binom{n}{k}B^k\,\mathbb{E}_f\left[\varepsilon^{n-k}\right]$$
$$+ \frac{1}{\eta}\frac{\lambda}{2}\sum_{l=0}^{n-1}\sum_{m=0}^{\min\{n-l,n-1\}}\binom{n}{n-l-m,l,m}A^{n-l-m}\mathbb{E}_f\left[\varepsilon^l\right]\mathbb{E}_f\left[\varepsilon^m\right]$$

[6] The assumption that an acquired firm is replaced by a new entrant is commonly used to keep the total measure of firms constant. See, for example, Celik et al. (2022), among others.

and, in particular, the first two moments of the target firms' product quality distribution are

$$\mathbb{E}_f[\varepsilon] = \frac{1}{\eta}\left[\left(\eta + \frac{\lambda}{2}\right)\mathbb{E}_g[\varepsilon] + \frac{\lambda}{2}A + RB\right] \tag{4}$$

and

$$\mathbb{E}_f[\varepsilon^2] = \frac{1}{\eta^3}\left[\eta^2\left(\left(\eta + \frac{\lambda}{2}\right)\mathbb{E}_g[\varepsilon^2] - \frac{\lambda}{2}A^2\right) + \lambda\left(\left(\eta + \frac{\lambda}{2}\right)(\mathbb{E}_g[\varepsilon] + A) + RB\right)^2 + BR\eta\left((2\eta+\lambda)\mathbb{E}_g[\varepsilon] + \lambda A + (\eta + 2R)B\right)\right], \tag{5}$$

respectively.

2.2. A frictional goods market with market power

In this section, we present the trading side of our model economy. We borrow many of the model assumptions from Rubinstein and Wolinsky (1985) and Choi and Rocheteau (2024).[7] However, we endogenize the product quality distribution by introducing M&A dynamics, while Choi and Rocheteau (2024) present a model of market power by taking as given the firms' product quality distribution.

Environment. There are two types of goods in the economy. First, there is a continuum of products whose qualities are seller-specific. These perishable products are produced by sellers on the spot in bilateral meetings with buyers. Second, there is also a homogeneous perishable good, called *numéraire*, that all agents produce and consume.

At any point in time, there is a flow, $\omega > 0$, of homogeneous buyers into the economy. At rate $\alpha > 0$, a buyer is matched with a seller picked randomly and uniformly from the pool of all sellers. Upon meeting, the buyer and the seller engage in Kalai bargaining to determine how much of its product the seller will produce on the spot and how much numéraire will be transferred from the buyer to the seller. After meeting a seller, the buyer exits the economy regardless of the outcome of the bargaining. That is, trading is an option, and not a requirement, for a pair of buyer and seller, as is typical in these bilateral trade models.

Bilateral bargaining. In a meeting in which the seller produces y units of its product of quality ε and sells it to the buyer at a per-unit price of p, the buyer's surplus is $\varepsilon u(y) - py$ and the seller's surplus is $py - c(y)$.

We assume that the terms of trade (y, p) are determined by the proportional Kalai bargaining:

$$\max_{y,p} \varepsilon u(y) - py$$

s.t.

$$py - c(y) = \theta(\varepsilon u(y) - py),$$

where $\theta \in [0, \infty)$ is the seller's relative bargaining power (Kalai, 1977). Importantly, θ will be endogenously determined by a seller's rent-seeking activities (see Eq. (10)), and this mechanism is what generates an endogenous distribution of market power in our model as in Choi and Rocheteau (2024). For now, taking the bargaining power, θ, as given, the interior solution of the bargaining is as follows:

$$\varepsilon u'(y^*) = c'(y^*) \quad \text{and} \quad p = \frac{\theta}{\theta+1}\frac{\varepsilon u(y^*)}{y^*} + \frac{1}{\theta+1}\frac{c(y^*)}{y^*}.$$

Hence, the bargaining solution leads to a buyer's surplus of $\frac{1}{\theta+1}(\varepsilon u(y^*) - c(y^*))$, while the seller's surplus is $\frac{\theta}{\theta+1}(\varepsilon u(y^*) - c(y^*))$.

To be able to derive sellers' continuation utility in closed form, in what follows, we further assume that $u(y) = \sqrt{y}$ and that $c(y) = c_0 y$ for some $c_0 > 0$. Note that, *ceteris paribus*, these specifications imply a higher consumer surplus from a higher quality product, but firms bear the same production cost function.

Market power. With our functional form assumptions on the buyer's utility and the seller's cost, a meeting between the seller of ε-quality product and a buyer results in the exchange of $y^* = \frac{\varepsilon^2}{4c_0^2}$ units of the product for $p = \left(\frac{\theta(\varepsilon)}{\theta(\varepsilon)+1} + 1\right)c_0$ units of numéraire per unit, where we emphasize the dependence of $\theta(\varepsilon)$ on ε. The precise relation between the two will become clear shortly when we define and solve the seller's problem below.

Now we illustrate how this model generates various measures of market power by taking an arbitrary function $\theta(\varepsilon)$. At the firm-level, a most common measure of market power is to look at the markup. In our economy, the markup of the seller of product ε is

$$\mu(\varepsilon) \equiv \frac{p}{c'(y^*)} - 1 = \frac{\theta(\varepsilon)}{\theta(\varepsilon)+1}.$$

[7] See Section 3.2 and 6 of Choi and Rocheteau (2021).

From a market-wide perspective, a common measure of market power is the Herfindahl–Hirschman Index (HHI), defined as follows

$$HHI \equiv \int [s(\varepsilon)]^2 \, dF(\varepsilon), \tag{6}$$

where $s(\varepsilon)$ is the market share of the firm with product quality ε and equal to

$$s(\varepsilon) \equiv \frac{p(\varepsilon) y^*(\varepsilon)}{\int p(\varepsilon') y^*(\varepsilon') \, dF(\varepsilon')} = \frac{[\mu(\varepsilon)+1]\varepsilon^2}{\int [\mu(\varepsilon')+1](\varepsilon')^2 \, dF(\varepsilon')}. \tag{7}$$

Hence, both firms' markups and the market HHI will be determined once we solve for the endogenous bargaining power $\theta(\varepsilon)$.

Seller's problem. Now we are ready to describe the sellers' continuation utility. By assuming that sellers discount the future at rate $\rho > 0$, an application of the Bellman's principle of optimality implies that the continuation utility of a seller of product quality ε at steady state satisfies

$$\begin{aligned}\rho V(\varepsilon) = \max_{\theta \in [0,\infty)} \Bigg\{ & \underbrace{-v\left(\theta; \mathbb{E}_f\left[\varepsilon'\right]\right)}_{\text{rent-seeking cost}} + \underbrace{\alpha \frac{\omega}{\alpha} \frac{\theta}{\theta+1} \left(\varepsilon u\left(y^*\right) - c\left(y^*\right)\right)}_{\text{flow value of trading}} \\ & - \underbrace{\eta V(\varepsilon)}_{\text{exit}} + \underbrace{\tilde{R}\left(V(\varepsilon+B) - V(\varepsilon)\right)}_{\text{flow value of R\&D activities}} \\ & + \underbrace{\tilde{\lambda} \iint \frac{1}{2}\left[V\left(\varepsilon'+\varepsilon+A\right) + V\left(\varepsilon''\right) - V(\varepsilon) - V\left(\varepsilon'\right)\right] dF\left(\varepsilon'\right) dG\left(\varepsilon''\right) \Bigg\}}_{\text{flow value of an M\&A deal}}\end{aligned} \tag{8}$$

subject to

$$\varepsilon u'\left(y^*\right) = c'\left(y^*\right),$$

where $\tilde{R}$ and $\tilde{\lambda}$ denote the firm's R&D intensity and M&A intensity, respectively. The first term on the RHS of (8) is the flow cost of rent-seeking activities that support a seller bargaining power of θ. In writing this cost, we made it explicit that it depends on the aggregate productivity in the economy. This assumption captures the idea that, in an economy with more productive firms, increasing market power requires more investment in rent-seeking activities. The second term is the flow value of gains from trade, which is equal to the Poisson rate of meeting a buyer (α) times the steady-state mass of buyers (ω/α) times the seller's bargaining share ($\frac{\theta}{\theta+1}$) times the joint surplus of a trade between a buyer and the seller of product quality ε. The third term is the flow value of a death shock that arrives with the Poisson rate of η. The fourth term represents the flow value of R&D activities; that is, at Poisson rate $\tilde{R}$, the firm achieves a breakthrough and its product quality jumps from ε to $\varepsilon + B$. The last line is the flow value of an M&A deal, which is equal to the Poisson arrival rate of an M&A deal ($\tilde{\lambda}$) times the symmetric bargaining share (1/2) times the joint surplus of an M&A deal between the seller of quality ε and another seller of quality ε' that is picked randomly and uniformly from the steady-state distribution of sellers F. Upon meeting, these two sellers bargain over an acquisition price anticipating that the acquirer will leave the table with a certain product quality of $\varepsilon + \varepsilon' + A$ and the acquired firm will be replaced with a new firm with uncertain product quality of ε'' that comes from the distribution G. Assuming $\varepsilon \geq \varepsilon'$, the acquisition price of

$$P\left(\varepsilon, \varepsilon'\right) = \frac{1}{2} \int \left[V\left(\varepsilon'+\varepsilon+A\right) - V\left(\varepsilon''\right) - V(\varepsilon) + V\left(\varepsilon'\right)\right] dG\left(\varepsilon''\right)$$

convinces the firm of initial quality ε to acquire the firm of initial quality ε' by leaving both firms with same expected surplus of

$$V\left(\varepsilon'+\varepsilon+A\right) - V\left(\varepsilon'\right) + P\left(\varepsilon, \varepsilon'\right) = V\left(\varepsilon'+\varepsilon+A\right) - V(\varepsilon) - P\left(\varepsilon, \varepsilon'\right),$$

which is equal to the surplus stated in the second line of (8). Similarly, if $\varepsilon < \varepsilon'$, the acquisition price of

$$P\left(\varepsilon', \varepsilon\right) = \frac{1}{2} \int \left[V\left(\varepsilon'+\varepsilon+A\right) - V\left(\varepsilon''\right) + V(\varepsilon) - V\left(\varepsilon'\right)\right] dG\left(\varepsilon''\right)$$

convinces the firm of initial quality ε' to acquire the firm of initial quality ε by leaving both firms with same expected surplus of

$$V\left(\varepsilon'+\varepsilon+A\right) - V\left(\varepsilon'\right) - P\left(\varepsilon', \varepsilon\right) = V\left(\varepsilon'+\varepsilon+A\right) - V(\varepsilon) + P\left(\varepsilon', \varepsilon\right),$$

which is equal to the surplus stated in the second line of (8). Hence, the last line of the sellers' HJB Eq. (8) captures both cases in which the firm with product quality ε acts as the acquirer or is acquired.

With our maintained assumptions $u(y) = \sqrt{y}$ and $c(y) = c_0 y$ and by assuming a linear rent-seeking cost $v\left(\theta; \mathbb{E}_f\left[\varepsilon'\right]\right) = v_0 \left(\mathbb{E}_f\left[\varepsilon'\right]\right)^2 \theta$ for some $v_0 > 0$, (8) becomes

$$\begin{aligned}(\rho+\eta) V(\varepsilon) = \max_{\theta \in [0,\infty)} \Bigg\{ & -v_0 \left(\mathbb{E}_f\left[\varepsilon'\right]\right)^2 \theta + \omega \frac{\theta}{\theta+1} \frac{\varepsilon^2}{4c_0} + \tilde{R}\left(V(\varepsilon+B) - V(\varepsilon)\right) \\ & + \tilde{\lambda} \iint \frac{1}{2}\left[V\left(\varepsilon'+\varepsilon+A\right) + V\left(\varepsilon''\right) - V(\varepsilon) - V\left(\varepsilon'\right)\right] dF\left(\varepsilon'\right) dG\left(\varepsilon''\right) \Bigg\}.\end{aligned} \tag{9}$$

Then, provided that ε is sufficiently large, the sellers' first-order condition implies that the endogenous bargaining power for firms of type ε is

$$\theta(\varepsilon) = \sqrt{\frac{\omega}{4c_0 v_0}} \frac{\varepsilon}{\mathbb{E}_f[\varepsilon']} - 1, \tag{10}$$

which, in turn, implies a seller's endogenous markup of

$$\mu(\varepsilon) = \frac{\theta(\varepsilon)}{\theta(\varepsilon)+1} = \frac{\sqrt{\frac{\omega}{4c_0 v_0}} \frac{\varepsilon}{\mathbb{E}_f[\varepsilon']} - 1}{\sqrt{\frac{\omega}{4c_0 v_0}} \frac{\varepsilon}{\mathbb{E}_f[\varepsilon']}}. \tag{11}$$

We therefore have the following proposition characterizing market power in the steady state.

Proposition 3 (*Steady-State Markups and Market Concentration*)**.** *At the steady state, the endogenous markups, market shares, and market concentrations are*

$$\mu(\varepsilon) = 1 - \frac{\mathbb{E}_f[\varepsilon']}{\varepsilon}\sqrt{\frac{4c_0 v_0}{\omega}},$$

$$s(\varepsilon) = \frac{2\varepsilon^2 - \sqrt{\frac{4c_0 v_0}{\omega}}\mathbb{E}_f[\varepsilon']\varepsilon}{2\mathbb{E}_f[\varepsilon^2] - \sqrt{\frac{4c_0 v_0}{\omega}}\left(\mathbb{E}_f[\varepsilon]\right)^2}, \text{ and}$$

$$HHI = \frac{4\mathbb{E}_f[\varepsilon^4] - 4\sqrt{\frac{4c_0 v_0}{\omega}}\mathbb{E}_f[\varepsilon^3]\mathbb{E}_f[\varepsilon'] + \frac{4c_0 v_0}{\omega}\mathbb{E}_f[\varepsilon^2]\left(\mathbb{E}_f[\varepsilon']\right)^2}{\left(2\mathbb{E}_f[\varepsilon^2] - \sqrt{\frac{4c_0 v_0}{\omega}}\left(\mathbb{E}_f[\varepsilon]\right)^2\right)^2},$$

respectively.

This proposition provides us with intuitive comparative statics. In the cross section of firms, those with higher product quality are able to charge a higher markup to buyers. This is because of endogenous rent seeking: a seller of high quality product generates a large trade surplus, and so, she has incentive to devote resources to rent seeking to be able to have a large bargaining share. In addition, the overall level of markups is larger in an economy with frequent arrival of buyers, low cost of rent seeking, and low marginal cost of production. In all of these cases, the expected trade surplus is large and sellers want to capture a large share of that large "pie".

We now solve for the equilibrium value function. Using (11), (9) becomes

$$\begin{aligned}(\rho+\eta)V(\varepsilon) = &\left(\sqrt{\frac{\omega}{4c_0}}\varepsilon - \sqrt{v_0}\mathbb{E}_f[\varepsilon']\right)^2 + \tilde{R}\left(V(\varepsilon+B) - V(\varepsilon)\right) \\ &+ \tilde{\lambda}\iint \frac{1}{2}\left[V(\varepsilon'+\varepsilon+A) + V(\varepsilon'') - V(\varepsilon) - V(\varepsilon')\right] dF(\varepsilon')\, dG(\varepsilon'').\end{aligned} \tag{12}$$

The unique quadratic solution of (12) for a given M&A search intensity $\tilde{\lambda}$ and R&D intensity $\tilde{R}$ is obtained by conjecturing a quadratic solution for V and matching the coefficients.

Proposition 4 (*Equilibrium Value Function*)**.** *Let* $K = \sqrt{\frac{\omega}{c_0}}$*. Assume the supports of the distributions* F *and* G *are bounded below such that all* $\varepsilon \geq \frac{2\sqrt{v_0}}{K}\mathbb{E}_f[\varepsilon]$*. Then, the unique quadratic solution to the sellers' HJB Eq. (9) is*

$$V(\varepsilon) = V_0 + \frac{K}{4(\rho+\eta)}\left[\left(-4\sqrt{v_0}\mathbb{E}_f[\varepsilon] + \frac{K}{\rho+\eta}\left(2\tilde{R}B + \tilde{\lambda}\mathbb{E}_f[\varepsilon+A]\right)\right)\varepsilon + K\varepsilon^2\right], \tag{13}$$

where

$$\begin{aligned}(\rho+\eta)V_0 = &\, v_0\left(\mathbb{E}_f[\varepsilon]\right)^2 + \frac{K}{8(\rho+\eta)}\tilde{\lambda}\Bigg[-4\sqrt{v_0}\mathbb{E}_f[\varepsilon]\mathbb{E}_g[\varepsilon+A] \\ &+ \frac{K}{\rho+\eta}\tilde{\lambda}\mathbb{E}_f[\varepsilon+A]\mathbb{E}_g[\varepsilon+A] + K\mathbb{E}_g[\varepsilon^2] + KA\left(2\mathbb{E}_f[\varepsilon]+A\right)\Bigg] \\ &+ \tilde{R}B\frac{K}{4(\rho+\eta)}\left[-4\sqrt{v_0}\mathbb{E}_f[\varepsilon] + \frac{K}{\rho+\eta}\left(2\tilde{R}B + \tilde{\lambda}\mathbb{E}_f[\varepsilon+A]\right) + KB\right] \\ &+ \tilde{R}\tilde{\lambda}\frac{K^2}{4(\rho+\eta)^2}B\mathbb{E}_g[\varepsilon+A].\end{aligned}$$

The main benefit of Proposition 4 is that it provides a clear analytical characterization of the dynamic value for a seller of sustaining a particular M&A effort level $\tilde{\lambda}$ and R&D effort level $\tilde{R}$.

2.3. Endogenous M&A and R&D efforts

We now endogenize a seller's choice of $\tilde{\lambda}$ and $\tilde{R}$ by taking as given the cross-sectional average of firms' product quality, $\mathbb{E}_f[\varepsilon]$, which itself depends on the aggregate M&A effort level, λ, and the aggregate R&D effort level, R, of other firms as given by Proposition 2.

More precisely, we now allow entrant sellers to commit to a level of M&A efforts, $\tilde{\lambda}$, and a level of R&D efforts, $\tilde{R}$, before the realization of their individual state ε, which realizes randomly from the exogenous distribution G. This is in the spirit of the endogenous-search-intensity over-the-counter (OTC) trading models, e.g., Farboodi et al. (2023) and Üslü and Weill (2026). In addition, entrant sellers are *ex ante* identical in the endowment, m, of the numéraire that they can allocate across the M&A and the R&D activities. This translates into the following constrained optimization problem:

$$\max_{\tilde{\lambda},\tilde{R}\in\mathbb{R}_+} \ (\rho+\eta)\,\mathbb{E}_g[V(\varepsilon)]$$

subject to

$$\chi_0\tilde{\lambda}+r_0\tilde{R}\leq m,$$

where $\chi_0>0$ and $r_0>0$ are the marginal costs of increasing the M&A and the R&D efforts, respectively.[8] By using Eq. (13), one easily sees that the objective is increasing in both $\tilde{\lambda}$ and $\tilde{R}$. This guarantees that the constraint binds at an optimum. Then, by substituting

$$\tilde{R}=\frac{m-\chi_0\tilde{\lambda}}{r_0}$$

into the objective, one can write the first-order condition and second-order condition for an interior solution:

$$\begin{aligned}\tilde{\lambda}=&\left[2K\left(\mathbb{E}_f[\varepsilon+A]-2B\frac{\chi_0}{r_0}\right)\left(2B\frac{\chi_0}{r_0}-\mathbb{E}_g[\varepsilon+A]\right)\right]^{-1}\\&\times\left[(\rho+\eta)\left(K\Xi+4\sqrt{v_0}\mathbb{E}_f[\varepsilon]\left(2B\frac{\chi_0}{r_0}-\mathbb{E}_g[\varepsilon+A]\right)-2KB\frac{\chi_0}{r_0}\mathbb{E}_g[2\varepsilon+B]\right)\right.\\&\left.+2K\frac{m}{r_0}B\left(\mathbb{E}_f[\varepsilon+A]+\mathbb{E}_g[\varepsilon+A]-4B\frac{\chi_0}{r_0}\right)\right]\end{aligned}\tag{14}$$

and

$$\left(\mathbb{E}_f[\varepsilon+A]-2B\frac{\chi_0}{r_0}\right)\left(\mathbb{E}_g[\varepsilon+A]-2B\frac{\chi_0}{r_0}\right)<0,$$

respectively, where

$$\Xi=\mathbb{E}_g\left[(\varepsilon+A)^2\right]+2\mathbb{E}_f[\varepsilon]\,\mathbb{E}_g[\varepsilon+A].$$

Note that the second-order condition effectively puts an upper bound and a lower bound on χ_0/r_0 so that an interior solution obtains. That is, if this ratio was too high (resp. too low), entrant sellers would only engage in R&D (resp. M&A) activities. Because, in practice, firms do both M&A and R&D, we will only focus on cases where the second-order condition holds.

Assuming an interior solution, the first-order condition (14), together with Eq. (4), is also the sellers' best-response function in choosing their M&A search efforts, $\tilde{\lambda}(\lambda)$. This best-response function reveals that the game among the entrant sellers is a game with strategic complementarity in actions, in certain region of the parameter space. Proposition 5 provides a sufficient condition for the strategic complementarity to obtain.

Proposition 5 (*Strategic Complementarity*)**.** *If m is sufficiently large, $\frac{d\tilde{\lambda}(\lambda)}{d\lambda}>0$.*

The intuition is as follows. Eq. (4) shows that the cross-sectional average of firms' product quality is an increasing function of the aggregate M&A intensity. And, when the cross-sectional average of firms' product quality is higher, an entrant's marginal benefit from increasing her own M&A intensity is higher, which makes $\tilde{\lambda}(\cdot)$ an increasing function of λ in (14) resulting in the mentioned strategic complementarity, provided that m is sufficiently large. Note that this is different from the game among entrant investors in the models of Farboodi et al. (2023) and Üslü and Weill (2026), which always exhibit strategic substitutability. The reason is that in those models, when aggregate search intensity is higher, the equilibrium level of misallocation is lower, which lowers the gains from trade, and in turn, lowers the marginal benefit of choosing a higher search intensity. While this situation generates unique equilibrium in those search-based OTC trading environments, our model with strategic complementarity shows that search-based M&A environments are prone to multiple equilibria.

[8] This constrained optimization problem essentially assumes that firms face a trade-off between investing in R&D and engaging in M&A activity: at any interior solution, greater investment in R&D implies less investment in M&A, and vice versa. Seru (2014) provides empirical support for this assumption in the context of conglomerate mergers, showing that acquiring firms allocate fewer resources to internal R&D.

Proposition 6 (*Equilibrium Search Intensities*). *Assume*

$$\frac{m}{r_0}B > \left(\eta + \frac{m}{2\chi_0}\right)\left(2B\frac{\chi_0}{r_0} - \mathbb{E}_g\left[\varepsilon + A\right]\right) > 0.$$

Let

$$C = 2B\frac{\chi_0}{r_0} - \mathbb{E}_g\left[\varepsilon + A\right],$$

$$a_2 = Kr_0^2C^2,$$

$$a_1 = r_0C\left\{K\left(2C\eta r_0 - 3Bm\right) - (\rho+\eta)\, r_0\left(K\mathbb{E}_g[\varepsilon + A] + 2\sqrt{v_0}C\right)\right\},$$

$$\begin{aligned} a_0 = {} & 2B^2K\, m^2 - 4BCK\,\eta m r_0 \\ & + (\rho+\eta)\left\{\eta r_0^2\left[\,\mathbb{E}_g[(\varepsilon+A)^2] - 2CK\mathbb{E}_g[\varepsilon] - 4C(A+C)\sqrt{v_0}\,\right]\right. \\ & \left. + 2B\chi_0\,\eta r_0\left[4C\sqrt{v_0} - BK\right] + 2Bmr_0\left[K\,\mathbb{E}_g[\varepsilon + A] + 2\sqrt{v_0}C\right]\right\}, \end{aligned}$$

and

$$\Delta = a_1^2 - 4a_0a_2.$$

Define the function $H : \mathbb{R} \to \mathbb{R}$ *such that*

$$H(\lambda) = a_0 + a_1\lambda + a_2\lambda^2.$$

(i) ***(Unique symmetric equilibrium)*** *If* $H(0)\, H\left(\frac{m}{\chi_0}\right) < 0$ *holds, then there exists a unique symmetric equilibrium. The equilibrium M&A intensity and R&D intensity are*

$$\lambda = \frac{-a_1 - \operatorname{sgn}(H(0))\sqrt{\Delta}}{2a_2}$$

and

$$R = \frac{m - \chi_0\lambda}{r_0}.$$

respectively.

(ii) ***(Two symmetric equilibria)*** *If* $H(0) > 0$, $H\left(\frac{m}{\chi_0}\right) > 0$, $H\left(\frac{-a_1}{2a_2}\right) < 0$, *and* $\frac{-a_1}{2a_2} \in \left(0, \frac{m}{\chi_0}\right)$ *hold, there exist exactly two symmetric equilibria. The equilibrium M&A intensities* $\lambda_1 < \lambda_2$ *are*

$$\lambda_{1,2} = \frac{-a_1 \mp \sqrt{\Delta}}{2a_2},$$

with the associated R&D intensities

$$R_i = \frac{m - \chi_0\lambda_i}{r_0}$$

for $i \in \{1, 2\}$.

The intuition for Proposition 6 is the following. Under the modeling assumptions, the fixed-point problem for the best response to other firms' search intensities reduces to finding the roots of a quadratic polynomial. Two equilibria exists when there is an incentive for a single firm to deviate and search more when no other firm is searching for $M\&A$, $H(0) > 0$, and when there is also an incentive to deviate and search more when all firms are devoting the maximal amount of resources to $M\&A$ search, $H\left(\frac{m}{\chi_0}\right) > 0$. In addition, there must be intermediate region where firms have an incentive to deviate and search less.

Proposition 6 shows that an equilibrium with low M&A and high R&D activity can coexist with another equilibrium featuring high M&A and low R&D activity. This raises four natural questions. First, which of these equilibria is stable? Second, in which equilibrium are markups higher? Third, which equilibrium delivers higher product quality, consumer surplus, or social welfare? Fourth, for empirically plausible parameter values, does the model fall in the region with a unique equilibrium or with multiple equilibria?

Let us start with the stability analysis. The stability of the equilibrium depends on the sign of the first derivative of the best-response function evaluated at these equilibria. An interior equilibrium λ is stable if and only if $\tilde{\lambda}'(\lambda) < 1$. Proposition 7 proves that, when there are two equilibria, only the equilibrium with lower λ is stable, provided that our sufficient condition for interior λ holds.

Proposition 7 (*Stability*). *Assume*

$$\frac{m}{r_0}B > \left(\eta + \frac{m}{2\chi_0}\right)\left(2B\frac{\chi_0}{r_0} - \mathbb{E}_g\left[\varepsilon + A\right]\right) > 0.$$

In the parameter region where there exist exactly two symmetric equilibria with the M&A intensities $\lambda_1 < \lambda_2$, the equilibrium associated with λ_1 is stable whereas the equilibrium associated with λ_2 is unstable.

Propositions 6 and 7 are interesting due to their potential empirical and regulatory implications. Suppose that empirically plausible parameter values imply multiple equilibria.[9] Suppose that the economy is initially at the unstable coordination equilibrium, λ_2, characterized by high M&A and low R&D activity. If regulators were concerned about this combination and sufficiently tightened antitrust policy—mapped in our model to an increase in χ_0—the economy would transition toward the stable equilibrium, λ_1, with lower M&A and higher R&D. Whether such a shift improves social welfare, or even consumer welfare, is not immediate. *Ceteris paribus,* both M&A and R&D raise product qualities while also affecting market power and increasing markups, so the welfare consequences hinge on their relative magnitudes. Thus, a careful welfare accounting would be necessary to decide whether moving from the λ_2-equilibrium to the λ_1-equilibrium is indeed desirable.

In the end, if the regulators decided that the tightened antitrust policy was a bad idea, they could revert it. However, it does not guarantee that the economy would return to original coordination equilibrium, because the equilibrium with λ_1 is stable. That is, the economy may remain trapped in the low-M&A equilibrium, even if regulators come to view that outcome as undesirable. Hence, our model with M&A search complementarities questions the extent to which unintended consequences of some antitrust policy changes may successfully be reversed.

2.4. Welfare

Next, we turn to the welfare analysis. At this point, it is not clear whether the equilibrium with the lower M&A effort λ_1 would have higher social welfare, as several forces in the model operate with opposite effects on welfare. This equilibrium is characterized by lower M&A but higher R&D activities, relative to the λ_2-equilibrium. Therefore, it is not clear in which equilibrium product qualities are better. On the one hand, higher product qualities would imply higher consumer and producer surpluses in bilateral trade between buyers and sellers, as well as higher labor earnings, and in turn, tend to increase social welfare. On the other hand, the equilibrium with higher product qualities would be associated with more investment in rent-seeking activities, which reduces social welfare. To determine which effect dominates, we conduct a careful welfare analysis.

We start by defining the flow social welfare as

$$\rho\mathcal{W} = \int \left(-v\left[\theta\left(\varepsilon\right)\right] + \omega \left\{ \max_{y} \varepsilon u\left(y\right) - c\left(y\right) \right\} + \omega c_0 y^*\left(\varepsilon\right) \right) dF\left(\varepsilon\right),$$

where $y^*\left(\varepsilon\right) = \arg\max_{y} \varepsilon u\left(y\right) - c\left(y\right)$. The first term inside the integral is the flow cost of rent-seeking activities by a seller of product quality ε. The second term is the flow gains from trade. Note that there is no coefficient $\theta\left(\varepsilon\right)$ inside the flow gains from trade because the social planner considers the shares of the surplus that accrue to the buyer $\left(1/\left(\theta\left(\varepsilon\right)+1\right)\right)$ as well as to the seller $\left(\theta\left(\varepsilon\right)/\left(\theta\left(\varepsilon\right)+1\right)\right)$, which sums up to 1. The last term inside the integral is the earnings of the hand-to-mouth workers who make the production possible. Thus, the integral stands in for the aggregation of rent-seeking costs and gains from trades including labor earnings across all product-quality types according to their steady-state measure F.

By using our main assumptions regarding utility, $u\left(y\right) = \sqrt{y}$, production costs, $c\left(y\right) = c_0 y$, and rent-seeking costs, $v\left(\theta\right) = v_0\left(\mathbb{E}_f\left[\theta'\right]\right)^2\theta$, the flow social welfare is

$$\rho\mathcal{W} = \int \left(-v_0\left(\mathbb{E}_f\left[\theta'\right]\right)^2\theta\left(\varepsilon\right) + \frac{\omega}{4c_0}\varepsilon^2 + \frac{\omega}{4c_0}\varepsilon^2 \right) dF\left(\varepsilon\right).$$

Finally, by using the sellers' optimal choice of $\theta\left(\varepsilon\right)$s, we obtain the following proposition.

Proposition 8 (*Equilibrium Welfare Given λ*)**.** *Let $K = \sqrt{\frac{\omega}{c_0}}$. Assume the supports of the distributions F and G are bounded below such that all $\varepsilon \geq \frac{2\sqrt{v_0}}{K}\mathbb{E}_f\left[\theta\right]$. Then, the equilibrium welfare is given by*

$$\mathcal{W}\left(\lambda\right) = \frac{1}{\rho}\left(\underbrace{-\left(\frac{K\sqrt{v_0}}{2} - v_0\right)\left(\mathbb{E}_f\left[\varepsilon\right]\right)^2}_{\text{Cost of market power}} + \underbrace{\frac{K^2}{2}\mathbb{E}_f\left[\varepsilon^2\right]}_{\text{Trading surplus}} \right),$$

where $\mathbb{E}_f\left[\varepsilon\right]$ and $\mathbb{E}_f\left[\varepsilon^2\right]$ are given by (4) and (5), respectively.

The first part of the welfare, $-\frac{1}{\rho}\left(\frac{K\sqrt{v_0}}{2} - v_0\right)\left(\mathbb{E}_f\left[\varepsilon\right]\right)^2 < 0$, is the pure welfare cost of market power. In an equilibrium with high average product quality, the welfare cost of market power is larger, i.e., it is more negative. This is because firms selling high-quality products are able to charge larger markups. However, in an equilibrium with high product quality, the welfare gain from trading surpluses and labor earnings, $\frac{1}{\rho}\left(\frac{K^2}{2}\mathbb{E}_f\left[\varepsilon^2\right]\right)$, is higher as well. Importantly, note that trading surplus is the sum of the producer's

[9] Our calibrated numerical example in Section 3.1 suggests that this is the case.

surplus and the consumer's surplus. In that sense, markups also make a *positive* contribution to social welfare. Overall, our theoretical model with explicit formulae for the first two moments of the product quality distribution provides us with a framework to weigh the opposing welfare effects of M&A activities against each other. In the next section, we calibrate the model using empirically plausible parameter values and show that markups are positive at the constrained-efficient allocation,[10] because the planner, when maximizing welfare subject to the model's constraints, balances the benefits of market power against its costs.

3. Quantification and counterfactuals

3.1. Numerical example

We now provide quantitative implications of our model by using numerical simulations, as in Choi and Rocheteau (2024) and Cavenaile et al. (2021). In particular, we follow David (2021), to calibrate parameters that capture the main features of the firm size distribution. We therefore set the exit shock, η, to 6.5% and the discount rate, ρ, to 4%. Following Cavenaile et al. (2021), we target the M&A efficiency gains, A, to 0.0146, and the innovation step size, B, to 0.0115.[11] We also set m and r_0 to 0.0000434 and 0.0000512, respectively, so that total entry costs and the average firm's R&D expenditures are consistent with the entry-cost scale and R&D-cost scale calibrations in Cavenaile et al. (2021). The last five parameters are the average product quality of the entrant firms, $\mathbb{E}_g[\varepsilon] = 0.0021$; the product quality variance of the entrant firms, $\mathbb{V}_g[\varepsilon] = 0.0008243$; the marginal M&A cost, $\chi_0 = 0.0001874$; the reduced-form parameter $K = 0.3677$, which combines operating costs and the flow of entrants; and the rent-seeking cost, $\upsilon_0 = 0.0002045$.

With the described parameter choices, the stable interior equilibrium of our model (i.e., the equilibrium with λ_1) matches five of the empirical moments reported by Cavenaile et al. (2021): the effective merger arrival rate ($\lambda_1 = 0.1625$), average net markup ($\mathbb{E}_f[\mu(\varepsilon)] = 0.3498$), variance of markups ($\mathbb{V}_f[\mu(\varepsilon)] = 0.12$), average profitability (14.4%),[12] and the ratio of R&D expenditures to GDP (2.43%). We see this simple calibration as a first step towards bringing the model to the data, with the main goal of providing mostly a qualitative assessment of the forces in the model.

The calibrated parameters imply that the economy satisfies the sufficient conditions for two symmetric equilibria described in part ii) of Proposition 6. Fig. 2 illustrates the relation between the equilibrium M&A intensities and the endogenous market structure. The stable low M&A search intensity equilibrium is associated with higher R&D activities, higher average product quality, and higher markups. Because average product quality has a first order and markups have a second order effect on consumer welfare, consumer welfare is higher in the low M&A intensity equilibrium. In that sense, consumers face a trade-off between product quality and the level of markups across the two equilibria. Our calibration implies that consumers are better off in this stable equilibrium with lower search intensity.

Our numerical result that consumers face a trade-off between product quality and markup levels relates to the recent of work of Menzio (2026), who argue that interpreting the markups through the lens of the monopolistic competition models could lead to incorrect policy recommendations. In Menzio (2026), markups are efficient. In contrast, while the level of markups is not efficient in our model, they have efficiency-enhancing properties. That is why social welfare and consumer welfare are higher in the high average markup equilibrium than those in the low average markup equilibrium. Because the product quality distribution is better (in the sense of first-order stochastic dominance) in the high markup equilibrium, consumers are on net better off even though they end up paying a higher markup for firms' products.

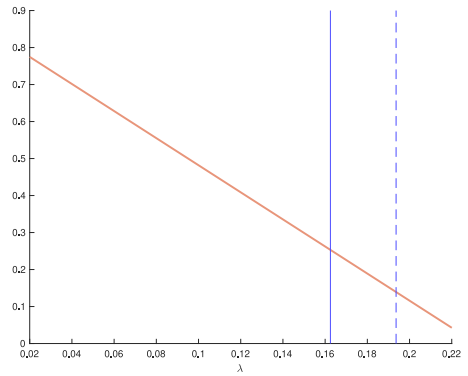
Another lesson from Fig. 2 is that the two common empirical measures of market power, average markup and HHI, do not necessarily move in unison. While average markup would imply that the low M&A intensity equilibrium is associated with greater market power, HHI would imply the opposite. In the high M&A intensity equilibrium, investment in R&D is lower, which in turn reduces average product quality. Lower average product quality makes rent-seeking less costly. As a result, more productive firms increase their markups by more relative to the rest of the economy, leading to a more concentrated market in terms of sales revenue, as reflected in a higher HHI.

Finally, Fig. 2 can be used to characterize the numerical solution to a constrained planner's problem. Suppose the planner cannot affect trade between consumers and firms or firms' continuous rent-seeking efforts, and is therefore constrained in that sense. However, the planner can influence how entrant firms allocate their resources across M&A and R&D activities. If the planner maximizes social welfare, it chooses $\lambda^* = 0.0306$ and $R^* = 0.736$. Thus, in both of the privately optimal equilibria, firms invest too much in M&A and too little in R&D. The reason is that, whenever the conditions for multiple equilibria are satisfied, investment in M&A exhibits strategic complementarities, which in turn crowd out firms' investment in R&D.

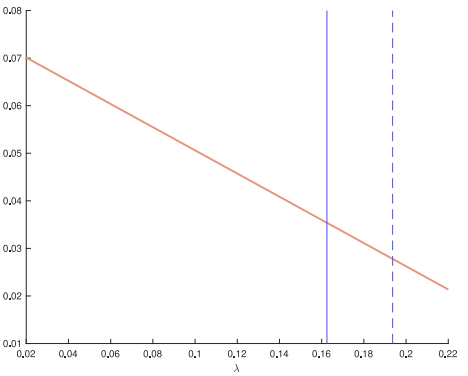
[10] See Fig. 2 and the surrounding discussion.

[11] More precisely, Cavenaile et al. (2021) assume multiplicative functional forms for post-M&A and post-R&D productivity, whereas we adopt additive functional forms in order to use characteristic function techniques. Accordingly, in our calibration we choose A and B so that the average firm's productivity gains under our specification are consistent with the relative gain parameters in Cavenaile et al. (2021).

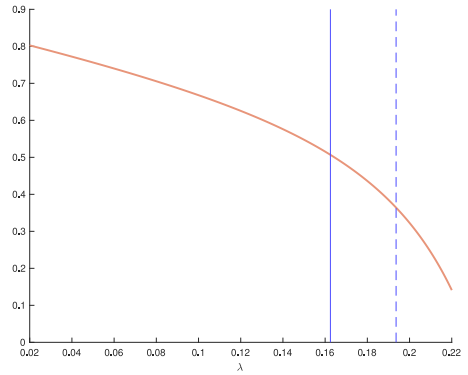
[12] Following Cavenaile et al. (2021), we define the empirical profitability ratio as the ratio of operating income before depreciation to sales. Its counterpart in our model is equal to the ratio of static profit flow after rent-seeking cost to sales.



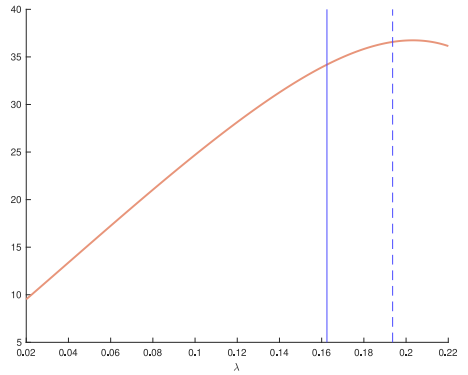
(a) R&D Intensity



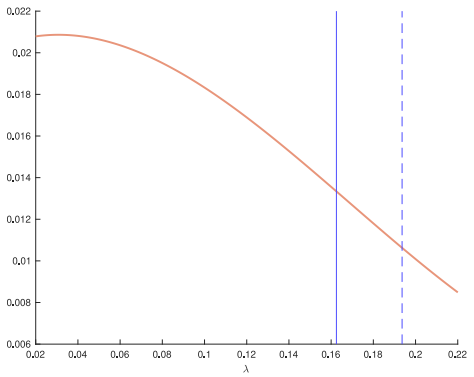
(b) Average Quality



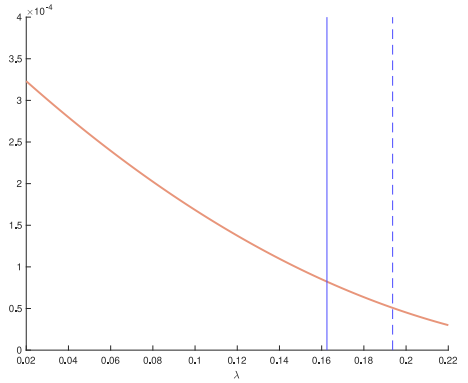
(c) Average Mark-up



(d) HHI



(e) Welfare



(f) Consumer Welfare

Fig. 2. Impact of M&A Search Intensity.
Note: The figures plot the changes of the endogenous outcomes as all firms change their choice of M&A search intensity by staying on their budget line, $\chi_0\lambda + r_0 R = m$. The solid (dashed) vertical line represents the equilibrium with low (high)M&A search intensity, λ_1 (λ_2).

3.2. Empirical and policy implications

M&A search intensities are the key drivers of the model's outcomes, as seen in the previous section. We now study the impact of exogenous changes in the model's parameters on equilibrium M&A search intensities. While our framework does not model antitrust policies explicitly, we interpret the magnitude of the M&A search costs (χ_0) as an indicator of the stringency of antitrust policies. There are several ways in which more stringent antitrust policies increase the costs of concluding an acquisition, for instance by

requiring more detailed and lengthier pre-merger filing. More stringent merger reviews can also lead to costly break-up fees. While only a small fraction of the proposed deals require a Second Request, unsuccessful mergers can be very costly.[13] For instance, the acquisition of T-Mobile USA by AT&T that was ultimately blocked by the DOJ led to a breakup fee of around $4.2 billion (Fidrmuc et al., 2018).

Fig. 3 shows the comparative statics for the two interior equilibrium search intensities, λ_1 and λ_2, with respect to some key exogenous parameters. The low M&A equilibrium λ_1 is the only stable interior equilibrium and its comparative statics typically follow from direct effects, and so, are straightforward to interpret. The high M&A equilibrium λ_2 is an unstable equilibrium. That is, it is best understood as a self-fulfilling prophecy sustained thanks to the M&A search complementarities. For example, Panel (c) shows that λ_1 decreases with the M&A search cost χ_0. When it is more costly to search for M&A partners, firms search less leading to a lower equilibrium search intensity λ_1. However, λ_2 increases with χ_0. This high search intensity λ_2 is optimal for a firm only because other firms are choosing this high search intensity λ_2, generating large M&A surpluses. If the search cost χ_0 is larger, firms would collectively need to choose an even higher search intensity to generate even larger M&A surpluses so that the complementarities would outweigh the increase in χ_0. That is why higher χ_0 leads to a higher equilibrium search intensity λ_2.

Another example is Panel (d), which shows that λ_1 increases with the M&A synergy benefit A. A higher A implies a larger M&A surplus. In turn, firms search more leading to a higher equilibrium search intensity λ_1. However, λ_2 decreases with A. Again, the intuition follows from the self-fulfilling nature of this equilibrium. This high search intensity λ_2 is optimal for a firm only because other firms are choosing this high search intensity λ_2, generating large M&A surpluses. Now, a higher A implies that it is possible to generate the same large surpluses by choosing a lower search intensity. That is why higher synergy benefit A leads to a lower equilibrium search intensity λ_2. Most other comparative statics of λ_1 and λ_2 also follow from the same logic, and so, they exhibit opposite signs of slopes in each of those panels as in Panel (c) and (d).

One virtue of our model is that it offers a new lens through which to interpret historical merger waves in the United States. In particular, the model suggests that merger activity need not move one-for-one with slowly evolving fundamentals. Instead, when the economy admits multiple equilibria, the same underlying environment may be consistent with both a low-merger regime and a high-merger regime. In usual times, firms coordinate on the low-merger equilibrium, λ_1, in which acquisition activity is limited and stable. In that regime, firms devote a larger share of resources to organic growth rather than acquisitions, and merger activity remains contained even if the underlying gains from matching are positive. This provides a natural benchmark for periods in which aggregate merger volume is modest and fluctuations in M&A activity appear subdued.

The model also helps rationalize why merger waves can appear sudden, persistent, and seemingly disproportionate relative to the size of the underlying fundamental shocks. A sufficiently favorable change in fundamentals — for example, improved financing conditions, technological complementarities, regulatory permissiveness, or sectoral reorganization — may increase the attractiveness of M&A. But when strategic complementarities are strong, fundamentals alone may not be the whole story. A favorable shock, together with a coordination device or "sunspot", may enable firms to move away from the low-merger equilibrium and coordinate on the high-merger equilibrium associated with elevated acquisition activity. In this sense, the model speaks to episodes in which merger booms seem to feed on themselves: once firms expect rivals to acquire aggressively, the private return to doing so rises, which further reinforces merger activity. The resulting amplification mechanism can help explain why historical merger waves often display a clustered and self-reinforcing character rather than a smooth response to fundamentals.

At the same time, the high-merger regime is fragile in our framework. The equilibrium with unusually high merger intensity is unstable, so it should be interpreted not as a permanently sustainable state, but rather as a precarious configuration that can persist only so long as coordination on it is maintained. This fragility offers a natural interpretation of why merger waves eventually end, often abruptly. A destabilizing shock — such as tighter credit conditions, a shift in antitrust enforcement, disappointing post-merger outcomes, or a broader deterioration in expectations — can break the coordination sustaining the high-merger regime and push the economy back toward the low-merger equilibrium. Seen through this perspective, merger waves are not merely periods in which firms become exogenously more acquisitive; they are episodes in which favorable fundamentals and coordination interact to produce temporarily elevated M&A activity, followed by an eventual reversion once that coordination becomes untenable.

Fig. 4 illustrates how M&A search costs, χ_0, and R&D costs, r_0, affect the conditions for multiple equilibria and, hence, the possibility of a merger wave to occur. Proposition 6 states that multiple equilibria arise when $H(0) > 0$ and $H(m/\chi_0) > 0$.[14] Fig. 4 shows that these conditions are satisfied when χ_0 is sufficiently low. This suggests a natural mechanism through which a merger wave may come to an end: a tightening of antitrust policy that causes an increase in χ_0 and moves the economy outside of the region where the high-equilibrium exists. Consistent with this interpretation, Cho and Chung (2022) note that "the Great Merger Wave", which began at the end of the 19th century, lasted only about a decade and ended abruptly following stronger enforcement of the Sherman Antitrust Act. Similarly, Jarrell et al. (1988) argue that relatively weak state-level anti-takeover regulation contributed to the emergence of the fourth merger wave, which occurred from 1984 to 1989.

Fig. 4 further shows that the conditions for multiple equilibria are met when r_0 is sufficiently high, namely when R&D investment is sufficiently costly. This implies that, in addition to conventional antitrust policy, an alternative way to prevent or curtail merger waves is to subsidize firms' R&D activity. In the model, policies such as R&D subsidies or tax credits operate by lowering r_0, thereby reducing the scope for multiple equilibria.

[13] "Just to step back, in any given year, we see up to 3000 merger filings that get reported to us. Around 2% of those actually get a second look by the government, so you have 98% of all deals that, for the most part, are going through". https://techcrunch.com/2024/06/15/ftc-chair-lina-khan-on-startups-scaling-and-innovations-in-potential-law-breaking/

[14] Together with $H\left(-\frac{a_1}{2a_2}\right) < 0$, which holds for all values of χ_0 and r_0 shown in the figure.

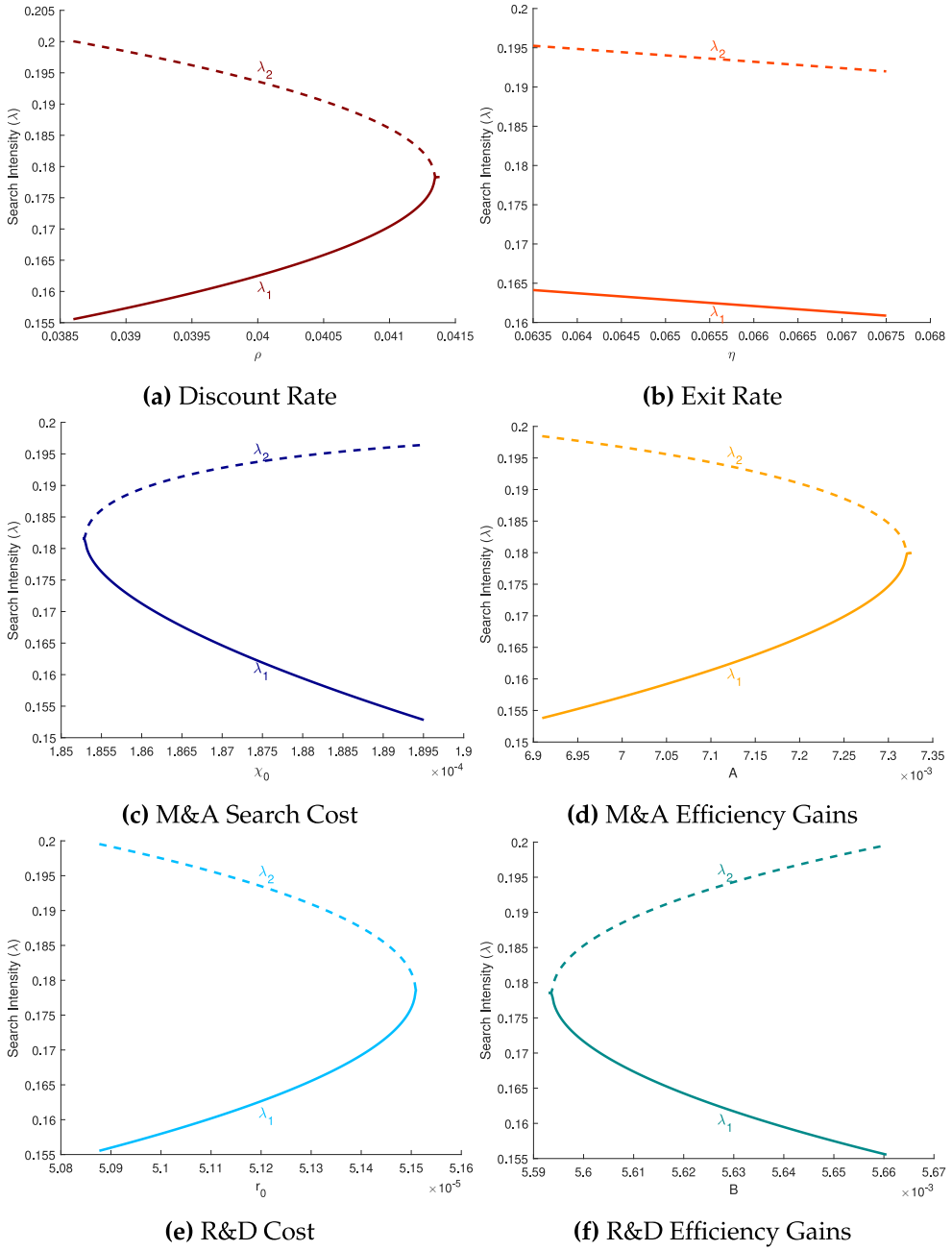
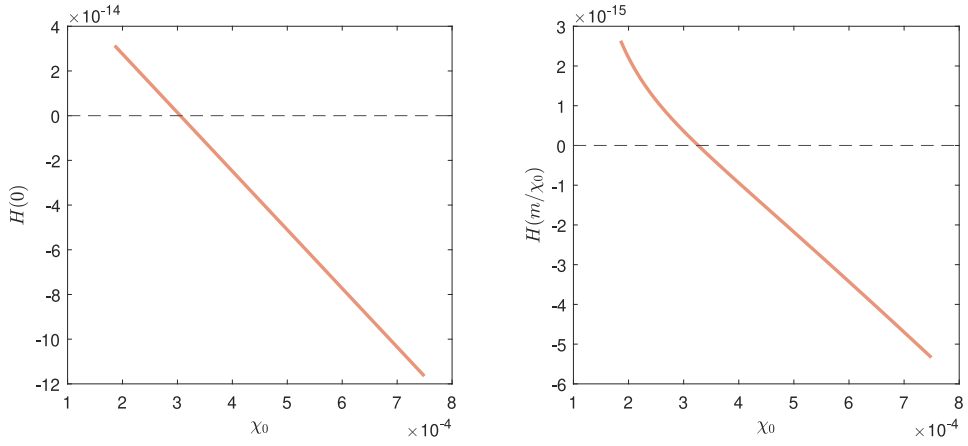


Fig. 3. Impact of parameter changes on M&A intensity.
Note: Each panel shows the impact of exogenous changes in the model's parameters on the two candidate equilibrium search intensities, λ_1, the stable equilibrium, (solid line) and λ_2, the unstable equilibrium (dashed line).

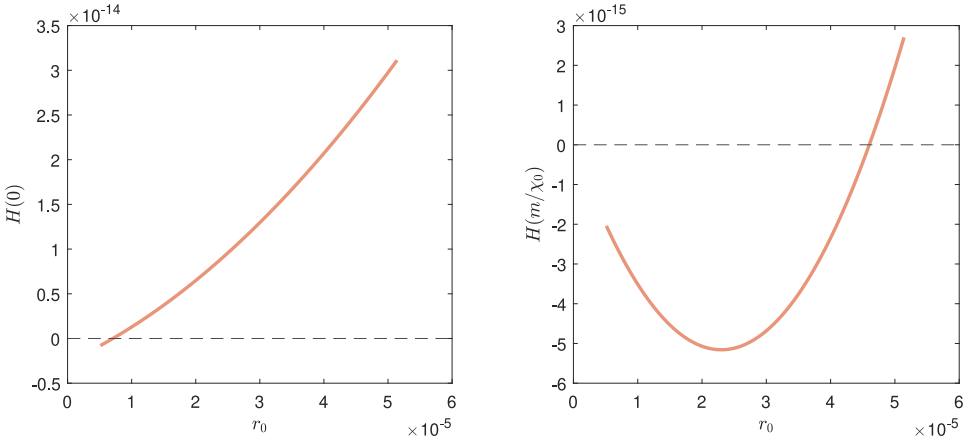
4. Conclusion

We develop a search-based theory of mergers and acquisitions in which heterogeneous firms allocate resources between external growth through acquisitions and internal growth through R&D. In the model, product-market power is endogenous, and the firm distribution evolves through exit, innovation, and mergers. The framework is tractable enough to characterize the stationary distribution analytically and to solve for firms' equilibrium M&A and R&D choices.

Our main result is that strategic complementarities in acquisition incentives can generate multiple equilibria. In particular, a stable equilibrium with low M&A and high R&D can coexist with an unstable equilibrium with high M&A and low R&D. This



(a) Impact of M&A search cost (χ_0)



(b) Impact of R&D cost (r_0)

Fig. 4. Impact of parameter changes on the conditions for multiple equilibria.
Note: Charts on the left show the change in the incentives for searching when no other firm searches, $H(0)$, and the charts on the right the incentives for searching when all other firms devote the maximum amount of resources in search, $H\left(m/\chi_0\right)$. Parameter choices except for χ_0 and r_0 are the same as in the baseline calibration. All values of χ_0 and r_0 on the grids satisfy $\Delta > 0$ and $H\left(-\frac{a_1}{2a_2}\right) < 0$ so that at least one interior equilibrium exists.

mechanism offers a simple way to think about merger waves as episodes in which favorable fundamentals and coordination jointly sustain temporarily elevated acquisition activity. The quantitative analysis also shows that equilibrium differences in merger activity matter for innovation, product quality, markups, concentration, and welfare. In our calibrated example, the stable equilibrium is associated with higher R&D, higher average product quality, higher average markups, and higher consumer welfare. More broadly, the model implies that antitrust policy and innovation policy can affect not only the level of merger activity, but also the set of equilibria the economy can sustain.

The paper also points to two natural directions for future research. First, our analysis focuses on a stationary equilibrium in which there is no aggregate growth. A natural next step is therefore to develop a version of the framework with a balanced growth path,[15] together with richer institutional features that would allow the model to speak more directly to real-world antitrust policy. Such an environment would likely require a fully numerical solution and is left for future work. Second, it would be valuable to extend the model to include aggregate uncertainty and stochastic equilibria. In such a setting, one could characterize equilibrium switching probabilities and study how the likelihood that a merger wave begins or ends depends on the aggregate state of the economy. This would provide a natural way to connect the theory more closely to the timing and fragility of historical merger waves.

[15] See Rocheteau (2025) for a recent example of a decentralized trade model with rich firm heterogeneity and balanced growth path.

Overall, the paper suggests that the effects of merger policy are inherently dynamic. Because firms choose between acquisitions and innovation, and because these choices interact strategically, policy can shape not only market structure at a point in time, but also the broader evolution of the economy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Mathematical appendix

Proof of Proposition 1

We first derive the characteristic function of the second term on the RHS of (1):

$$\int_{-\infty}^{\infty} Rf(\varepsilon - B)\, e^{iz\varepsilon} d\varepsilon = R\int_{-\infty}^{\infty} f(\varepsilon - B)\, e^{iz\varepsilon} d\varepsilon$$

$$= Re^{izB}\int_{-\infty}^{\infty} f(\varepsilon - B)\, e^{iz(\varepsilon - B)} d(\varepsilon - B) = Re^{iBz}\int_{-\infty}^{\infty} f(\varepsilon')\, e^{iz\varepsilon'} d\varepsilon' = Re^{iBz}\hat{f}(z).$$

Next, we derive the characteristic function of the last term on the RHS of (1):

$$\int_{-\infty}^{\infty}\left[\frac{\lambda}{2}\int_{-\infty}^{\infty} f(\varepsilon')\, f(\varepsilon - A - \varepsilon')\, d\varepsilon'\right] e^{iz\varepsilon} d\varepsilon = \frac{\lambda}{2}\int_{-\infty}^{\infty}\left[f(\varepsilon')\int_{-\infty}^{\infty} f(\varepsilon - A - \varepsilon')\, e^{iz\varepsilon} d\varepsilon\right] d\varepsilon'$$

$$= \frac{\lambda}{2}\int_{-\infty}^{\infty}\left[f(\varepsilon')\int_{-\infty}^{\infty} f(\varepsilon - A - \varepsilon')\, e^{iz(\varepsilon - A - \varepsilon')} d(\varepsilon - A - \varepsilon')\right] e^{iz(A+\varepsilon')} d\varepsilon'$$

$$= \frac{\lambda}{2} e^{iAz}\int_{-\infty}^{\infty}\left[f(\varepsilon')\,\hat{f}(z)\right] e^{iz\varepsilon'} d\varepsilon' = \frac{\lambda}{2} e^{iAz}\hat{f}(z)\int_{-\infty}^{\infty} f(\varepsilon')\, e^{iz\varepsilon'} d\varepsilon' = \frac{\lambda}{2} e^{iAz}\left(\hat{f}(z)\right)^2.$$

Using the linearity and differentiability of the characteristic function, one can obtain the characteristic functions of all other terms in (1), which leads to (2). By equating the LHS of (2) to zero, we obtain a quadratic equation for $\hat{f}(z)$. Its negative root gives the steady-state characteristic function (3). Its positive root does not lead to a characteristic function because it violates the fact that $\hat{f}(0) = 1$.

Proof of Proposition 2

To characterize the nth moment of the steady state distribution we use the KFE taken at steady state. We have

$$\left(\eta + \lambda - R\left(e^{iBz} - 1\right)\right)\hat{f}(z) = \left(\eta + \frac{\lambda}{2}\right)\hat{g}(z) + \frac{\lambda}{2} e^{iAz}\left(\hat{f}(z)\right)^2. \tag{15}$$

Apply the generalized Leibniz rule to the last term on the LHS:

$$\frac{d^n}{dz^n}\left[-R\left(e^{iBz} - 1\right)\hat{f}(z)\right] = -R\sum_{k+l=n}\binom{n}{k,l}(iB)^k e^{iBz}\hat{f}^{(l)}(z) + R\hat{f}^{(n)}(z)$$

$$= -Re^{iBz}\sum_{k=1}^{n}\binom{n}{k}(iB)^k\hat{f}^{(n-k)}(z) - R\left(e^{iBz} - 1\right)\hat{f}^{(n)}(z).$$

Hence, evaluating at $z = 0$,

$$\frac{d^n}{dz^n}\left[-R\left(e^{iBz} - 1\right)\hat{f}(z)\right]\Big|_{z=0} = -R\sum_{k=1}^{n}\binom{n}{k}(iB)^k\hat{f}^{(n-k)}(0).$$

Finally, multiplying by i^{-n} and using $i^{-m}\hat{f}^{(m)}(0) = \mathbb{E}_f[\varepsilon^m]$, we obtain

$$i^{-n}\frac{d^n}{dz^n}\left[-R\left(e^{iBz} - 1\right)\hat{f}(z)\right]\Big|_{z=0} = -R\sum_{k=1}^{n}\binom{n}{k}B^k\,\mathbb{E}_f[\varepsilon^{n-k}].$$

Now apply the generalized Leibniz rule to the last term on the RHS of (15):

$$\frac{d}{dz^n}\left[\frac{\lambda}{2} e^{iAz}\left(\hat{f}(z)\right)^2\right] = \frac{\lambda}{2} e^{iAz}\sum_{k+l+m=n}\binom{n}{k,l,m} i^k A^k\hat{f}^{(l)}(z)\,\hat{f}^{(m)}(z)$$

$$= \frac{\lambda}{2} e^{iAz} \sum_{l=0}^{n} \sum_{m=0}^{n-l} \binom{n}{n-l-m, l, m} i^{n-l-m} A^{n-l-m} \hat{f}^{(l)}(z) \hat{f}^{(m)}(z)$$

$$= i^n \frac{\lambda}{2} e^{iAz} \sum_{l=0}^{n} \sum_{m=0}^{n-l} \binom{n}{n-l-m, l, m} A^{n-l-m} i^{-l} \hat{f}^{(l)}(z) i^{-m} \hat{f}^{(m)}(z)$$

Now rewrite the double summation by singling out the two terms corresponding to $(l,m) \in \{(n,0),(0,n)\}$ and for which the corresponding multinomial term $\binom{n}{0,n,0} = 1$ such that

$$\frac{d}{dz^n}\left[\frac{\lambda}{2} e^{iAz} \left(\hat{f}(z)\right)^2\right] = i^n \frac{\lambda}{2} e^{iAz} \left(i^{-n} \hat{f}^{(n)}(z) \hat{f}(z) + \hat{f}(z) i^{-n} \hat{f}^{(n)}(z)\right)$$
$$+ i^n \frac{\lambda}{2} e^{iAz} \sum_{l=0}^{n-1} \sum_{m=0}^{\min\{n-l,n-1\}} \binom{n}{n-l-m, l, m} A^{n-l-m} i^{-l} \hat{f}^{(l)}(z) i^{-m} \hat{f}^{(m)}(z).$$

Finally, because by property of the characteristic function $i^{-n} \hat{f}^{(n)}(0) = \mathbb{E}_f[\varepsilon^n]$, then taking the nth derivative, multiplying by i^{-n}, and evaluating at $z = 0$ Eq. (15) we have

$$\eta \mathbb{E}_f[\varepsilon^n] = \left(\eta + \frac{\lambda}{2}\right) \mathbb{E}_g[\varepsilon^n] + R \sum_{k=1}^{n} \binom{n}{k} B^k \mathbb{E}_f[\varepsilon^{n-k}]$$
$$+ \frac{\lambda}{2} \sum_{l=0}^{n-1} \sum_{m=0}^{\min\{n-l,n-1\}} \binom{n}{n-l-m, l, m} A^{n-l-m} \mathbb{E}_f[\varepsilon^l] \mathbb{E}_f[\varepsilon^m];$$

therefore expressing the nth moment of f in terms of the nth moment of g and strictly lower order moments of f. It is clear than $\mathbb{E}_f[\varepsilon^1]$ is strictly increasing in A, λ, B, and R. It then follows by induction that every moment of f is strictly increasing with A, λ, B, and R.

Proof of Proposition 3

The first equation obtains by rearranging (11). The second equation obtains by plugging (11) into (7). Then, the second equation, together with (6), implies the last equation.

Proof of Proposition 4

We first conjecture a solution of the form

$$V(\varepsilon) = V_0 + V_1 \varepsilon + V_2 \varepsilon^2$$

for V_0, V_1, and V_2 to be determined. Then, by plugging this conjecture into the buyer's HJB (Eq. (12)) and matching coefficients, we obtain

$$V_0 = \frac{v_0 \left(\mathbb{E}_f[\varepsilon]\right)^2}{\rho + \eta} + \frac{K}{8(\rho+\eta)^2} \tilde{\lambda} \Big[-4\sqrt{v_0} \mathbb{E}_f[\varepsilon] \mathbb{E}_g[\varepsilon + A]$$
$$+ \frac{K}{(\rho+\eta)^2} \tilde{\lambda} \mathbb{E}_f[\varepsilon + A] \mathbb{E}_g[\varepsilon + A] + K \mathbb{E}_g[\varepsilon^2] + KA\left(2\mathbb{E}_f[\varepsilon] + A\right) \Big]$$
$$+ \tilde{R} B \frac{K}{4(\rho+\eta)^2} \left[-4\sqrt{v_0} \mathbb{E}_f[\varepsilon] + \frac{K}{\rho+\eta} \left(2\tilde{R}B + \tilde{\lambda} \mathbb{E}_f[\varepsilon + A]\right) + KB \right]$$
$$+ \tilde{R} \tilde{\lambda} \frac{K^2}{4(\rho+\eta)^3} B \mathbb{E}_g[\varepsilon + A],$$

$$V_1 = \frac{K}{4(\rho+\eta)} \left(-4\sqrt{v_0} \mathbb{E}_f[\varepsilon] + \frac{K}{\rho+\eta} \left(2\tilde{R}B + \tilde{\lambda} \mathbb{E}_f[\varepsilon + A]\right) \right),$$

and

$$V_2 = \frac{K^2}{4(\rho+\eta)},$$

which completes the proof.

Proof of Proposition 5

Let

$$T = 2B \frac{\chi_0}{r_0}.$$

Using (14), write $\tilde{\lambda}$ as

$$\tilde{\lambda} = \frac{N(m)}{D},$$

where

$$D \equiv 2K\Big(\mathbb{E}_f[\varepsilon + A] - T\Big)\Big(T - \mathbb{E}_g[\varepsilon + A]\Big)$$

and

$$N(m) \equiv (\rho + \eta)\Big[\big(K\Xi + 4\sqrt{v_0}\,\mathbb{E}_f[\varepsilon]\big)\big(T - \mathbb{E}_g[\varepsilon + A]\big) - 2KT\,\mathbb{E}_g[2\varepsilon + B]\Big] + 2K\frac{m}{r_0}B\Big(\mathbb{E}_f[\varepsilon + A] + \mathbb{E}_g[\varepsilon + A] - 2T\Big).$$

By assumption,

$$\Big(\mathbb{E}_f[\varepsilon + A] - T\Big)\Big(\mathbb{E}_g[\varepsilon + A] - T\Big) < 0.$$

Since

$$T - \mathbb{E}_g[\varepsilon + A] = -\Big(\mathbb{E}_g[\varepsilon + A] - T\Big),$$

we have

$$D = -2K\Big(\mathbb{E}_f[\varepsilon + A] - T\Big)\Big(\mathbb{E}_g[\varepsilon + A] - T\Big) > 0,$$

because $K > 0$.

Next, define

$$\tau \equiv \mathbb{E}_f[\varepsilon + A] + \mathbb{E}_g[\varepsilon + A] - 2T = \mathbb{E}_f[\varepsilon + A] + \mathbb{E}_g[\varepsilon + A] - 4B\frac{\chi_0}{r_0}.$$

Then the numerator can be written as

$$N(m) = N_0 + 2K\frac{m}{r_0}B\tau,$$

where

$$N_0 \equiv (\rho + \eta)\Big[\big(K\Xi + 4\sqrt{v_0}\,\mathbb{E}_f[\varepsilon]\big)\big(T - \mathbb{E}_g[\varepsilon + A]\big) - 2KT\,\mathbb{E}_g[2\varepsilon + B]\Big]$$

does not depend on m.

Under the additional assumptions $m > 0$ and $C > 0$, and using $K > 0$, $B > 0$, and $r_0 > 0$, the coefficient

$$2K\frac{B}{r_0}\tau$$

is strictly positive. Hence

$$N(m) \to +\infty \qquad \text{as } m \to \infty.$$

Therefore, there exists $\bar{m} > 0$ such that, for all $m > \bar{m}$,

$$N(m) > 0.$$

Since $D > 0$, it follows that, for all $m > \bar{m}$,

$$\tilde{\lambda} = \frac{N(m)}{D} > 0.$$

Finally, because $\lambda > 0$, we obtain

$$\frac{d\tilde{\lambda}(\lambda)}{d\lambda} > 0.$$

Therefore, if m is sufficiently large, then

$$\frac{d\tilde{\lambda}(\lambda)}{d\lambda} > 0.$$

Proof of Proposition 6

Fix parameters satisfying

$$\frac{m}{r_0}B > \left(\eta + \frac{m}{2\chi_0}\right)\left(2B\frac{\chi_0}{r_0} - \mathbb{E}_g\left[\varepsilon + A\right]\right) > 0. \tag{16}$$

Let $C \equiv 2B\chi_0/r_0 - \mathbb{E}_g[\varepsilon + A]$ so that $C > 0$ under (16), and note that $a_2 = Kr_0^2C^2 > 0$.

Step 1 (Reduction to a quadratic fixed-point equation). By construction of the best-response mapping $\tilde{\lambda}(\lambda)$ (defined in the text) and the binding budget line $R = (m - \chi_0\lambda)/r_0$, any symmetric equilibrium satisfies the fixed-point condition

$$\lambda = \tilde{\lambda}(\lambda).$$

Writing $\tilde{\lambda}(\lambda) = N(\lambda)/D(\lambda)$ with

$$D(\lambda) = \frac{2K}{\eta}\left[\frac{m}{r_0}B - \left(\eta + \frac{\lambda}{2}\right)C\right]C,$$

the fixed-point condition is equivalent to

$$N(\lambda) - \lambda D(\lambda) = 0, \qquad \text{and} \qquad D(\lambda) \neq 0.$$

Expanding $N(\lambda) - \lambda D(\lambda)$ and collecting powers of λ yields the quadratic polynomial

$$H(\lambda) = a_0 + a_1\lambda + a_2\lambda^2,$$

with coefficients (a_0, a_1, a_2) as stated in the proposition. Hence, any symmetric equilibrium intensity λ must satisfy $H(\lambda) = 0$, and the associated transfer is pinned down by the binding budget constraint as

$$R = \frac{m - \chi_0\lambda}{r_0}.$$

Step 2 (Well-definedness on the feasible interval). Define $\bar{\lambda} \equiv m/\chi_0$. Under (16) we have

$$\frac{m}{r_0}B > \left(\eta + \frac{\bar{\lambda}}{2}\right)C,$$

so for every $\lambda \in [0, \bar{\lambda}]$,

$$\frac{m}{r_0}B - \left(\eta + \frac{\lambda}{2}\right)C \;\geq\; \frac{m}{r_0}B - \left(\eta + \frac{\bar{\lambda}}{2}\right)C \;>\; 0.$$

Since $C > 0$, it follows that $D(\lambda) > 0$ on $[0, \bar{\lambda}]$, so the fixed-point equation $\lambda = \tilde{\lambda}(\lambda)$ is equivalent to $H(\lambda) = 0$ throughout the feasible set, with no spurious roots from clearing denominators.

Step 3 (Uniqueness under a sign change at the endpoints). Assume $H(0)H(\bar{\lambda}) < 0$. Since H is continuous, the Intermediate Value Theorem implies that H has at least one root in $(0, \bar{\lambda})$. Because $a_2 > 0$, H is a strictly convex quadratic, hence it can cross zero at most twice, and a sign change between $\lambda = 0$ and $\lambda = \bar{\lambda}$ implies there is exactly one root in $(0, \bar{\lambda})$. Moreover, with $\Delta = a_1^2 - 4a_2a_0 > 0$, the two (real) roots are

$$\lambda_{\pm} = \frac{-a_1 \pm \sqrt{\Delta}}{2a_2}, \qquad \lambda_- < \lambda_+.$$

The unique root in $(0, \bar{\lambda})$ is selected by

$$\lambda = \frac{-a_1 - \operatorname{sgn}(H(0))\sqrt{\Delta}}{2a_2},$$

as follows. If $H(0) = a_0 < 0$, then 0 lies between the two roots, so the unique positive root is λ_+; since $\operatorname{sgn}(H(0)) = -1$, the formula returns λ_+. If instead $H(0) > 0$ and $H(\bar{\lambda}) < 0$, then $\bar{\lambda}$ lies between the two roots and both roots are positive, so the unique root in $(0, \bar{\lambda})$ is λ_-; since $\operatorname{sgn}(H(0)) = +1$, the formula returns λ_-. The associated R is given by $R = (m - \chi_0\lambda)/r_0$.

Step 4 (Two equilibria when the minimum is negative and the vertex is interior). Assume $H(0) > 0$, $H(\bar{\lambda}) > 0$, and let $\lambda_v \equiv -a_1/(2a_2)$. If $\lambda_v \in (0, \bar{\lambda})$ and $H(\lambda_v) < 0$, then, because $a_2 > 0$, H attains its unique minimum at λ_v and is negative in a neighborhood of λ_v while remaining positive at the endpoints. Therefore H crosses zero exactly twice on $(0, \bar{\lambda})$, yielding exactly two distinct roots $\lambda_1 < \lambda_2$ in $(0, \bar{\lambda})$. Since $a_2 > 0$,

$$H(\lambda_v) = a_0 - \frac{a_1^2}{4a_2} = -\frac{\Delta}{4a_2},$$

so $H(\lambda_v) < 0$ implies $\Delta > 0$ and the two roots are given by the quadratic formula:

$$\lambda_{1,2} = \frac{-a_1 \mp \sqrt{\Delta}}{2a_2}.$$

The corresponding transfers are $R_i = (m - \chi_0\lambda_i)/r_0$ for $i \in \{1, 2\}$.

Combining Steps 1–4 establishes the claims in parts (i) and (ii).

Proof of Proposition 7

Let $\bar{\lambda} \equiv m/\chi_0$. Under

$$\frac{m}{r_0}B > \left(\eta + \frac{m}{2\chi_0}\right)\left(2B\frac{\chi_0}{r_0} - \mathbb{E}_g\left[\varepsilon + A\right]\right) > 0,$$

we have $C \equiv 2B\chi_0/r_0 - \mathbb{E}_g[\varepsilon + A] > 0$ and, for every $\lambda \in [0, \bar{\lambda}]$,

$$\frac{m}{r_0}B - \left(\eta + \frac{\lambda}{2}\right)C \;\geq\; \frac{m}{r_0}B - \left(\eta + \frac{\bar{\lambda}}{2}\right)C \;>\; 0.$$

Hence the denominator of the best-response map $\tilde{\lambda}(\lambda) = N(\lambda)/D(\lambda)$,

$$D(\lambda) = \frac{2K}{\eta}\left[\frac{m}{r_0}B - \left(\eta + \frac{\lambda}{2}\right)C\right]C,$$

satisfies $D(\lambda) > 0$ on $[0, \bar{\lambda}]$.

In the two-equilibrium region, (ii), of Proposition 6, the fixed-point condition $\lambda = \tilde{\lambda}(\lambda)$ is equivalent (on $[0, \bar{\lambda}]$) to the quadratic equation

$$H(\lambda) \equiv N(\lambda) - \lambda D(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 = 0,$$

with $a_2 = Kr_0^2C^2 > 0$. Let $\lambda_1 < \lambda_2$ denote the two distinct roots in $(0, \bar{\lambda})$.

Differentiate $H(\lambda) = N(\lambda) - \lambda D(\lambda)$ to obtain

$$H'(\lambda) = N'(\lambda) - D(\lambda) - \lambda D'(\lambda).$$

Moreover,

$$\tilde{\lambda}'(\lambda) = \frac{N'(\lambda)D(\lambda) - N(\lambda)D'(\lambda)}{D(\lambda)^2} = \frac{N'(\lambda) - \lambda D'(\lambda)}{D(\lambda)}.$$

At any fixed point λ_i we have $N(\lambda_i) = \lambda_i D(\lambda_i)$, and therefore

$$H'(\lambda_i) = \big(N'(\lambda_i) - \lambda_i D'(\lambda_i)\big) - D(\lambda_i) = D(\lambda_i)\big(\tilde{\lambda}'(\lambda_i) - 1\big).$$

Since $D(\lambda_i) > 0$, it follows that

$$\tilde{\lambda}'(\lambda_i) < 1 \iff H'(\lambda_i) < 0, \qquad \tilde{\lambda}'(\lambda_i) > 1 \iff H'(\lambda_i) > 0.$$

Because $H(\lambda)$ is a quadratic with $a_2 > 0$, it is strictly convex and its derivative is

$$H'(\lambda) = a_1 + 2a_2\lambda.$$

Let $\lambda_v \equiv -a_1/(2a_2)$ denote the unique minimizer. For a strictly convex quadratic, the minimizer lies strictly between the two distinct roots, so $\lambda_1 < \lambda_v < \lambda_2$. Hence

$$H'(\lambda_1) = 2a_2(\lambda_1 - \lambda_v) < 0, \qquad H'(\lambda_2) = 2a_2(\lambda_2 - \lambda_v) > 0.$$

Combining these sign inequalities with the equivalence above yields

$$\tilde{\lambda}'(\lambda_1) < 1 \quad \text{and} \quad \tilde{\lambda}'(\lambda_2) > 1.$$

Therefore λ_1 is stable and λ_2 is unstable.

Data availability

No data was used for the research described in the article.

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